2FiA: Towards WiFi Sensing-Based Authentication with Unique Biometrics

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***Abstract*—The emerging IEEE 802.11bf standard positions WiFi sensing as a key enabler of human biometric sensing for authentication. However, existing WiFi sensing-based authentication systems primarily depend on voluntary or semi-voluntary biometric traits, such as body motions or respiration. While effective, these traits lack strong uniqueness and are susceptible to imitation, making WiFi sensing systems vulnerable to behavioral mimicry attacks. In this paper, we introduce 2FiA, the first dual-biometrics WiFi sensing-based authentication system integrating semi-voluntary respiration and involuntary heartbeat. Respiration serves as the primary factor, offering efficient and continuous identity verification, while heartbeat reinforces system robustness through its highly unique and hard-to-imitate nature. For the first time, we propose a novel WiFi sensing pipeline to extract weak heartbeat signals by statistically localizing the thoracic region, effectively suppressing non-thoracic interference, and separating co-located respiration to refine the heartbeat signals. Furthermore, we leverage Respiratory Sinus Arrhythmia (RSA) as a physiological constraint to validate and enhance the extracted heartbeat signals by exploiting the intrinsic coupling between respiration and heartbeat. We then build the authentication model by integrating features from both respiration and heartbeat signals in a unified framework. Through extensive experiments involving 36 participants across diverse environments, we demonstrate that 2FiA delivers reliable authentication performance, particularly in multi-user and impersonation attack scenarios.**

1. Introduction

Biometric authentication has long leveraged unique physiological traits to verify identity, including fingerprints, facial features, iris patterns, retinal scans, palmprints, palm vein patterns, and voiceprints. These modalities are widely used due to their high distinctiveness, relative permanence, and strong resistance to forgery or imitation. However, most of them require dedicated sensing devices such as fingerprint readers, high-resolution cameras, or infrared scanners, limiting their scalability and seamless integration into everyday environments. With the emergence of IEEE 802.11bf [1],

TABLE 1: WiFi Sensing-based Authentication Systems

Biometric Trait	Technology	Uniqueness
Gait	XModal-ID [2], AutoID [3]	Low
Body Motions	3D-ID [4], Wi-Dist [5]	Low
Finger Gestures	Fingerpass [6], WiHF [7]	Low
Respiration	Cuvrb [8] , BreathID [9]	Medium
+ Heartbeat	**2FiA (Ours)**	**High**

the standardization of WiFi sensing marks a substantial step toward integrating authentication capabilities into widely deployed wireless infrastructure. Leveraging this existing, pervasive WiFi infrastructure helps overcome the hardware dependency and integration challenges faced by traditional biometric authentication systems.

However, existing WiFi sensing-based authentication systems, as shown in Table 1, rely on patterns embedded in voluntary or semi-voluntary biometric traits, such as gait [2], [3], body motions [4], [5], [10], [11], finger gestures [6], [7], and respiration [8], [9]. While these traits can effectively capture user-specific behavioral features, they are fundamentally different from traditional physiological traits like fingerprints or iris patterns, which are immutable and uniquely tied to an individual. In contrast, behavioral traits are more susceptible to spoofing, as they are observable and can be imitated by adversaries. For instance, an attacker can learn and imitate an individual's gait through prolonged observation or practice, such as walking on a treadmill with fine-tuned parameters [12]. Consequently, systems that rely solely on such traits are vulnerable to spoofing attacks. This underscores the need to augment systems with an additional unique biometric trait that is internally generated, involuntary, and inherently difficult to observe or imitate.

In this paper, we propose 2FiA, the first dual-biometrics WiFi sensing-based authentication system that combines semi-voluntary respiration and involuntary heartbeat biometric signals for enhanced authentication robustness and impersonation resistance. Inspired by the principle of multi-factor authentication [13], [14], 2FiA leverages respiration as the primary authentication factor due to its relatively strong and accessible signal, which enables efficient and continuous identity authentication [15], [16]. To further strengthen security, heartbeat is employed as a secondary factor, due to its high uniqueness and governed by the

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autonomic nervous system, which makes it difficult to consciously control and impersonate. This property enhances impersonation resistance when higher assurance is required. To realize this design, however, several technical challenges must be addressed:

Challenge 1: Localizing the Heartbeat Signal Source. Current WiFi-based authentication systems primarily focus on external and observable biometric signals such as gait, daily activities, finger gestures, and even respiration, which generate relatively strong WiFi signal reflections compared to surrounding environments. In contrast, the heartbeat is a subtle and internal signal that requires precise localization to be effectively extracted from a noisy environment. Traditional heartbeat sensing techniques, such as those based on millimeter-wave [17], [18], [19], [20], FMCW [21], [22], [23], [24], UWB [25], or Doppler radar [26], achieve this by directly sending signals toward the chest region. Unlike radar systems with highly directional beams, typical WiFi devices operate under non-directional or weakly directional conditions, making it challenging to localize a heartbeat source. To address this challenge, 2FiA first leverages respiration-enhanced energy patterns in 3D spectral space to approximate the thoracic region. It then refines this region by iteratively removing shared noises with adjacent regions, adaptively zooming in a reliable region containing hidden heartbeat signals.

Challenge 2: Separating Overlapped Respiration and Heartbeat Signals. After region refinement, respiration and heartbeat signals remain spatially overlapped, with respiration exhibiting much higher amplitude and dominating the signal space. This dominance of respiration in the signal space significantly overshadows the weaker heartbeat component. To overcome this challenge, 2FiA does not directly extract the weak heartbeat signal based on heartbeat-related frequencies [22]. Instead, it first focuses on accurately isolating the dominant respiratory component and then removes it to reveal the hidden real heartbeat signal. Given that respiration is non-stationary and exhibits time-varying frequency characteristics [27], which makes traditional bandpass filtering [28], [29] ineffective, we apply an adaptive noise decomposition to break the signal into Intrinsic Mode Functions (IMFs), each capturing distinct physiological patterns. The adaptive noise is scaled based on the signal strength in the thoracic region and adjacent regions to ensure robust decomposition across varying conditions. We then evaluate combinations of IMFs using the Breathing-to-Noise Ratio (BNR) [30], [31]. After identifying the respiration-related IMFs and residuals, we isolate them to subtract the dominant respiration component and noise, thereby revealing the true heartbeat signal candidate.

Challenge 3: Validating Heartbeat Signals through RSA. Although respiration and heartbeat signals have been separated, the extracted heartbeat signal remains weak and contains considerable residual noise. To ensure authentication reliability, 2FiA verifies candidate heartbeat signals by leveraging RSA [32], which is a natural pattern where heart rate increases during inhalation and decreases during exhalation. Using the extracted clean respiratory waveform, 2FiA segments breathing cycles and identifies inhalation and exhalation phases. It then applies autocorrelation to detect periodic heartbeat intervals and accepts only those whose temporal variation aligns with RSA-guided expectations. Specifically, the heartbeat intervals shorten during inhalation and lengthen during exhalation.

To realize robust authentication, we extract complementary morphological features from both respiration and heartbeat signals, each capturing distinct physiological characteristics. For respiration, we segment and interpolate each breathing cycle to represent its smooth, cyclic waveform using non-fiducial descriptors. For heartbeat, we focus on fiducial features by locating consistent peaks and valleys within each cardiac cycle. To enhance robustness against signal variability and noise, we further compute the first-order derivatives of both signals, revealing dynamic patterns such as instantaneous acceleration. These multi-scale features are then fused into a unified representation, which is fed into a Siamese Neural Network (SNN) with triplet loss. This approach enables learning discriminative feature representations essential for authentication.

The key contributions of this work are summarized as follows:

- We propose 2FiA, the first two-factor WiFi sensing based authentication system that combines semi-voluntary (respiration) and involuntary (heartbeat) biometric signals for enhanced authentication distinctiveness and impersonation resistance.
- We develop a novel signal processing pipeline that leverages WiFi sensing to extract subtle heartbeat signals by localizing the thoracic region, separating co-located interference to refine heartbeat signals, and validating them through RSA-based physiological correlation.
- We validate 2FiA through extensive experiments across diverse real-world environments and challenging conditions, including multi-user authentication and impersonation attacks. Experimental results show that 2FiA surpasses existing state-of-the-art WiFi respiration-based authentication systems, especially in multi-user scenarios and impersonation scenarios.

2. Models and Assumptions

In this section, we first introduce the system model and authentication workflow. Then, we define the threat model and assumptions.

2.1. System Model

2FiA leverages commodity WiFi devices to sense users' respiration and heartbeat signals, enabling contactless and seamless authentication. It operates in two distinct phases: an enrollment phase conducted in controlled conditions and an authentication phase deployed in complex real-world environments.

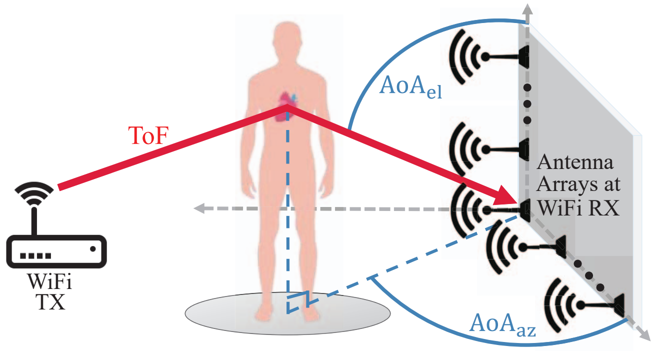


Figure 1: WiFi sensing model with 3D spatial modeling using AoA and ToF. The reflection path from the thoracic region is all we need.

Enrollment Phase: In the enrollment phase, 2FiA operates within a clean, interference-free indoor environment to establish a reliable biometric profile for each user. WiFi receivers capture reflected signals from multiple angles within 1 meter from user to the receiver, focusing on the thoracic region to extract high-fidelity respiration and heartbeat features. These features are processed into a user-specific representation and securely stored as an authentication reference. The controlled conditions ensure minimal multipath interference, enabling the system to accurately isolate and record core physiological traits.

Authentication Phase: In contrast, the authentication phase takes place in realistic and often dynamic environments, such as offices, classrooms, or shared living spaces. These settings introduce various sources of interference and environmental variability. In such scenarios, multiple users may be present simultaneously. Some of them are claimed users, while others may be co-present bystanders, all of whom can contribute additional human-induced reflections. Furthermore, even the legitimate users may perform unrelated activities during authentication, such as tapping, leg shaking, or hand waving, which introduce motion artifacts. Moreover, line-of-sight (LoS) signals and environmental multipaths from walls, floors, ceilings, and furniture further increase the complexity of signal interpretation. User locations may be placed outside the LoS path.

2.2. Threat Model and Assumptions

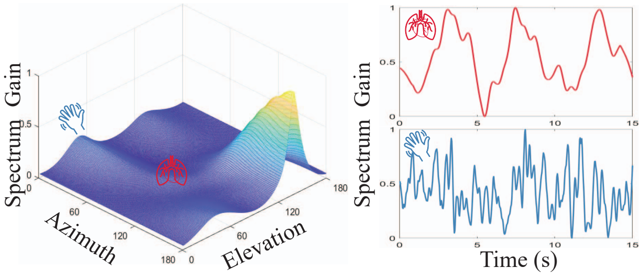
Our threat model considers adversaries aiming to deceive the WiFi sensing-based authentication system. We assume adversaries have the capability to position themselves within the effective WiFi sensing coverage area and intentionally manipulate reflected signals to attempt unauthorized access. Depending on the adversary's prior knowledge and techniques, we categorize the potential attacks into the following types:

Random Attack: In this attack scenario, the adversary lacks specific knowledge about the respiration and heartbeat patterns of legitimate users. Attackers simply enter the sensing range and produce random respiratory patterns or heartbeat rates, hoping that these arbitrary breathing and heartbeat patterns may accidentally bypass the authentication system.

Impersonation Attack: In this attack scenario, we assume the adversary can closely observe the legitimate user's breathing and acquire detailed knowledge of their respiratory characteristics. The attacker seeks to replicate the victim's breathing rate, amplitude, and posture to mimic these semi-voluntary patterns and deceive the system [33]. Specificially, to maximize attacker's impersonation capability, in our study, we deliberately selected attackers whose resting heart rates fell within a comparable range to the assigned victims. This design choice increased the likelihood that attackers could approximate the victim's physiological patterns. We then provided each attacker with the ground-truth breathing and heartbeat waveforms of their assigned victim, which they used as a reference to begin mimicking both signals. Note that the adversary can also have a similar body shape to the legitimate user.

Beyond respiration, an advanced adversary may also attempt to manipulate their own heartbeat signals to deceive the system. Specifically, breathing is semi-voluntary and mimicable, while heartbeat is involuntary and unalterable in pattern. However, heartbeat rate can be indirectly modulated through breathing (e.g., deep breathing slows it, faster breathing accelerates it), and the modulated heart rate can persist for a period of time. Based on this principle, attackers first adjusted their breathing to gradually align their heart rate with that of the victim as closely as possible. Once their heart rate had sufficiently approximated the target, they began mimicking the victim's breathing waveform patterns. That is how we achieved simultaneous breathing and heartbeat impersonation. While direct imitation of another person's heartbeat waveform is infeasible due to its involuntary and internal nature, attackers can indirectly influence their heart rate by altering their physical or emotional state. No modulation methods other than the aforementioned breathing modulation, such as exercise or stress induction, were considered in this study. The distinct morphological characteristics of each heartbeat cycle remain consistent regardless of heart rate. As a result, the effectiveness of such attacks is significantly constrained.

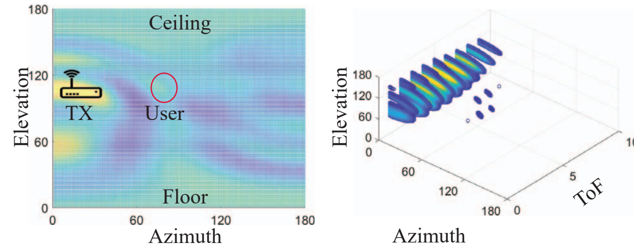
Replay Attack: In this scenario, the adversary captures and stores WiFi signals during a prior authentication session. Later, the adversary replays these recorded signals to the authentication system in an attempt to gain unauthorized access. We used a USRP to record and replay WiFi signals at different distances and angles relative to the receiver. While users may move or change posture, the environment—walls, furniture, and reflective surfaces—remains relatively static and forms a consistent spatial context. 2FiA leverages this spatial consistency as a feature: it constructs a 3D spatial spectrum based on spatial and temporal characteristics of each propagation path (see Section. 5). When the adversary records the signal from a different angle or position, their view of the environment is inherently different from that of the legitimate receiver. This mismatch in environmental perspective—what we call the environmental background



(a) Spatial spectrum highlights thoracic and non-thoracic regions.

(b) Temporal spectrum gain patterns mapped to the regions.

Figure 2: Illustration of spatial and temporal dynamics in a WiFi sensing environment. It demonstrates 2FiA's ability to physically separate target reflections region (thoracic region) and non-target reflections region (e.g., body parts region).



(a) 2D spatial spectrum

(b) 3D spatial spectrum.

Figure 3: Illustration of the advantage of 3D spatial profiling. By incorporating ToF, the 3D spectrum provides additional distance information [38], enabling precise localization of the thoracic reflection region.

signature—leads to spatial inconsistencies in the captured signal. 2FiA can detect and reject the replayed signal based on these spatial anomalies.

3. Preliminaries of WiFi Sensing

Modern WiFi systems, equipped with multiple-input multiple-output (MIMO) architectures and dense antenna arrays [34], have evolved beyond traditional communication to enable fine-grained sensing. These capabilities allow the detection of subtle biometric signals, such as heartbeat, manifesting as minute thoracic vibrations.

To extract such subtle biometric signals, we leverage the rich and multidimensional information encoded in WiFi Channel State Information (CSI). CSI characterizes how signals propagate and reflect within the environment, providing diverse features that can be analyzed to isolate propagation paths associated with these weak physiological vibrations.

We model the measured CSI matrix H as a set of spatio-temporal propagation paths. These paths are characterized by a steering vector matrix W and incident signal components F, corrupted by additive noise n [35], [36]. At specific time t, the resulting CSI channel is composed of a linear combination of multiple physical paths:

$$\begin{aligned} H &= WF + n \\ &= \underbrace{\sum_{c} W\, f(\theta^{c}_{az}, \theta^{c}_{el}, \tau_c)}_{\text{Thoracic Region Signals}} + \underbrace{\sum_{\hat{c}} W\, f(\theta^{\hat{c}}_{az}, \theta^{\hat{c}}_{el}, \tau_{\hat{c}})}_{\text{Non-Target Region Signals}} + n \end{aligned} \tag{1}$$

Here, F is the matrix of channel gains which encapsulates the incident signals arriving at the antenna array, including reflections induced by subtle physiological signals from the thoracic region and components from other non-thoracic sources. These non-thoracic components arise from various physical propagation paths, including the LoS path, environment-reflected paths, co-present human paths, and self-induced body multipath components.

To distinguish and isolate these signal components, we estimate the spatial and temporal characteristics of each propagation path by extracting several key parameters from the CSI, including:

Angle of Arrival (AoA) specifies the direction from which a WiFi signal reaches the receiver's antenna array [37]. As illustrated in Fig. 1, our three-dimensional (3D) sensing model represents AoA with two components: the azimuth angle (θ_{az}) in the horizontal plane and the elevation angle (θ_{el}) in the vertical plane. Given the L-shaped antennas array setup, the phase shift (Φ_{az} and Φ_{el}) is expressed as

$$\Phi_{az} = e^{-j2\pi d \cos(\theta_{az}) \sin(\theta_{el})/\lambda} \tag{2}$$

$$\Phi_{el} = e^{-j2\pi d \cos(\theta_{el})/\lambda} \tag{3}$$

where λ is the signal wavelength and d is the spacing between neighboring antennas (typically $\frac{\lambda}{2}$). The terms Φ_{az} and Φ_{el} capture the signal's spatial propagation along the azimuth and elevation axes, respectively.

Time of Flight (ToF) measures the propagation time of a WiFi signal from transmitter to receiver, enabling estimation of the path length to reflecting surfaces [36]. In WiFi systems, ToF is inferred from the phase shift (Φ_t) across subcarriers:

$$\Phi_t = e^{-j2\pi f_\delta \tau} \tag{4}$$

where f_δ is subcarrier spacing and τ is signal propagation delay.

4. System Overview

2FiA operates through a sensing-to-authentication pipeline that processes WiFi CSI to extract two types of biometric signals, respiration and heartbeat, and integrates them for two-factor authentication [39]. The system performs the following stages:

3D Spatial Modeling: The system begins by collecting CSI from a MIMO WiFi receiver. It then performs joint estimation of the AoA and ToF to construct a 3D spatial profile of the surrounding environment. This profile enables 2FiA to estimate and distinguish multiple signal propagation paths.

Spatial Localization of Subtle Heartbeat Signals: Motivated by the physically dominant patterns shown in Fig. 2, the system statistically analyzes spectrum gain to identify an initial thoracic region. Subsequently, contrastive noise cancellation is employed to eliminate shared noise across spatial neighboring reference points, enhancing the localization of heartbeat signal regions.

Heartbeat Signal Extraction and Verification: In the localized thoracic region, heartbeat signals remain co-located with other motions such as respiration. 2FiA first isolates the stronger respiration component using adaptive noise decomposition based on EEMD. By removing the respiration waveform and suppressing residual noise, the system reveals the underlying heartbeat signal. To further improve the quality of the extracted heartbeat signals, 2FiA leverages RAS as a physiological constraint, capturing the intrinsic coupling between respiration and heartbeat. It verifies each extracted heartbeat waveform against RSA-consistent timing variations, retaining only those signals that meet this criterion for authentication.

Feature Embedding and Authentication: Finally, 2FiA encodes the morphological and dynamic characteristics of both respiration and heartbeat waveforms into a unified feature representation, rather than simply concatenating the two patterns. This joint representation is embedded into a discriminative feature space using an SNN with triple loss. By comparing the embedding distances with registered user profiles in the feature space, 2FiA achieves a reliable authentication even in multi-user scenarios and against impersonation attacks."

5. 3D Spatial Modeling

Heartbeat induces subtle yet detectable vibrations on the chest surface, which in turn modulate the reflections of WiFi signals. Traditional heartbeat sensing technologies use directional radar to directly target the chest area and extract heartbeat signatures from the reflected signals [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. In contrast, WiFi operates under non-directional or weakly directional conditions, making the heartbeat more challenging to detect. To address this challenge, we leverage AoA in both the azimuth and elevation planes, along with ToF (detailed in Section 3), to construct a 3D spatial representation. These parameters jointly characterize the propagation paths of multipath components and are critical for isolating those modulated by heartbeat-induced vibrations.

We estimate the 3D spatial parameters by jointly resolving AoA and ToF using the MUSIC algorithm. The steering matrix W in Eqn. 1 is defined as:

$$\begin{aligned} W = \big[\, & \Phi^0_{az,el}, \Phi^0_{az,el}\Phi^1_t, \ldots, \Phi^0_{az,el}\Phi^{M-1}_t, \\ & \Phi^1_{az,el}, \Phi^1_{az,el}\Phi^1_t, \ldots, \Phi^1_{az,el}\Phi^{M-1}_t, \\ & \qquad \vdots \\ & \Phi^{N-1}_{az,el}, \Phi^{N-1}_{az,el}\Phi^1_t, \ldots, \Phi^{N-1}_{az,el}\Phi^{M-1}_t \,\big] \end{aligned} \tag{5}$$

where Φ_t represents the phase shift introduced by ToF across subcarriers, $\Phi_{az,el}$ denotes the phase shift by the L-shaped antenna arrays. Specifically, $\Phi_{az,el}$ can be calculated using either Φ_{az} represents the phase shift along the azimuth axis antenna array or Φ_{el} corresponds to the phase shift along the elevation axis antenna array. The total number of antennas N is the sum of antennas distributed along both the azimuth and elevation dimensions.

Under the assumption that noise and incident signals are uncorrelated, the covariance matrix of H decomposes as:

$$\overline{HH^*} = W\overline{FF^*}W^* + \overline{nn^*} \tag{6}$$

where $\overline{FF^*}$ represents the diagonal covariance matrix of incident signals, and nn^* denotes the noise covariance matrix.

Through eigenvalue decomposition of the covariance matrix, we obtain the noise subspace eigenvectors $\mathbf{E}_N$. The MUSIC algorithm leverages the orthogonality between signal and noise subspaces to compute the spatial power spectrum:

$$p(\theta_{az}, \theta_{el}, \tau) = \frac{1}{W^*\mathbf{E}_N\mathbf{E}_N^*W} \tag{7}$$

This 3D spectrum $p(\theta_{az}, \theta_{el}, \tau)$ provides an approximate estimate of the gain of $f(\theta_{az}, \theta_{el}, \tau)$, offering enhanced resolution for distinguishing physiological motion movements. Therefore, the 3D spectrum facilitates robust separation of thoracic-region reflections from environmental interferences through joint spatial–temporal processing, enabling reliable extraction of specific physiological motions.

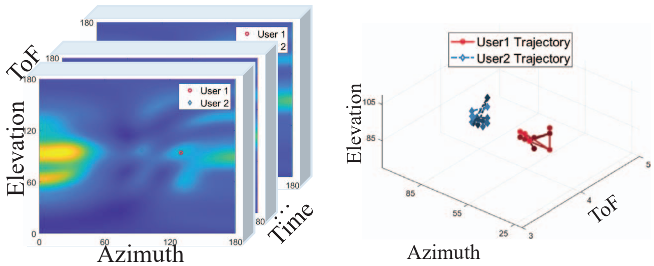
6. Spatial Localization of Subtle Heartbeat Signals

While the 3D spatial profiling framework provides high-resolution spatial information about signal distributions, it remains challenging to localize the thoracic region where both heartbeat-induced and respiration-induced vibrations originate. In this section, we introduce a localization method specifically designed to isolate the thoracic region where physiological signals originate. By focusing on the signal located in this region, we can mitigate surrounding interference, enabling the reliable extraction of heartbeat signals.

6.1. Initial Localization of the Thoracic Region

We first leverage respiration to localize the thoracic region, as respiratory movements generate stronger and more distinguishable signal variations than heartbeat, making them a reliable anchor for spatial initialization. Prior methods [40], [41] rely on tracking spectral peak trajectories, illustrated in Fig. 4, but often yield inconsistent patterns due to interference from environment reflectors and surrounding human activities, failing to capture the stable periodicity required for reliable authentication.

To overcome this limitation, our proposed approach does not depend on tracking trajectories but instead defines



(a) Time-series visualization of 3D spatial spectrums in 2FiA

(b) Trajectories of two users using the previous method.

Figure 4: Illustration of the limitations of traditional respiration sensing methods that track the highest energy peak. The resulting trajectories across multiple users are often out of range and discontinuous, lacking the regularity expected from periodic physiological motion. These inconsistencies demonstrate that relying solely on peak trajectories is insufficient for motion pattern extraction.

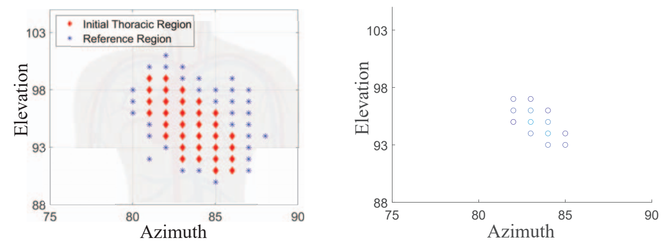
a baseline region encapsulating essential thoracic motion information. Through the 3D space modeling, this thoracic region effectively integrates key parameters, including Azimuth, Elevation, and ToF. Variations within this thoracic region implicitly carry crucial physiological signals. Because the estimated 3D space parameter power spectrum is highly related to the incident signals F, as demonstrated in Section 5. Focusing on this thoracic region significantly amplifies physiological signals.

To establish an initial region representing thoracic cavity, our method employs two primary steps:

Spatial Physiological Signal-to-Noise Ratio(S-PSNR) Calculation: The first step in our approach is the calculation of a PSNR (Physiological Signal-to-Noise Ratio) across the entire 3D spectrums in a short time window. Initially, S-PSNR is computed across the entire spectrum by comparing signal energy at respiratory and heartbeat frequencies against ambient noise levels [30]. We isolate the frequency band typically associated with human respiration and heartbeat and measure the energy at each frequency point within this band, contrasting it with surrounding noise energy averages. Unlike fixed region methods, our approach utilizes adaptive thresholding based on noise distribution, dynamically adjusting to environmental changes and selecting only points exhibiting significantly elevated S-PSNR.

$$\text{S-PSNR}(\theta_{az}, \theta_{el}, \tau) = \text{PSNR}(P(\theta_{az}, \theta_{el}, \tau)(t)) \quad (8)$$

Spatial Clustering: Subsequently, we apply spatial clustering to delineate baseline regions. High S-PSNR candidate cloud points identified during the previous step are grouped using a density-based clustering algorithm [42]. This clustering technique is advantageous as it does not necessitate predefined cluster counts and effectively identifies regions based on point density.



(a) Initial thoracic region via high S-PSNR clustering and reference region surrounding it.

(b) Refined thoracic region after contrastive noise signal cancellation.

Figure 5: Illustration of thoracic region localization.

6.2. Contrastive Isolation of the Thoracic Region

After initializing the thoracic region, we adopt an approach to refine and isolate the thoracic region, a smaller area within the broader thoracic region. This step involves both reference point selection and signal of reference points cancellation to enhance the weak heartbeat signals.

Reference Points Selection: We begin by identifying reference points that differ from the initialization region. These nearby points share similar motion disturbances (e.g., environmental noise, body motion, or respiration-induced displacement). Notably, the reference region is selected such that it does not contain localized cardiac motion. Unlike prior methods that leverage generic reference zones (e.g., between the neck and chest) [23], we adopt a spatially adaptive strategy that locates reference points adjacent to the initialization region. Specifically, we perform a breadth-first search (BFS) from each initialization point to identify the non-overlapping reference region.

Contrastive Noise Signal Cancellation: The key idea in this process is that the initialization region and its reference region often contain common noise, including spectrum-shared noise from the environmental and device-induced noise. This interference can be effectively modeled and removed using contrastive signal cancellation. Specifically, we extract time-series data from the 3D spatial spectrum by leveraging signal variations, forming:

Foreground : signals from the initialization region

Background : signals from the adjacent reference region

We then compute the contrastive covariance matrix [43]:

$$C_{contrastive} = (C_{Fore} - \alpha C_{back}) \quad (9)$$

where C_{Fore} and C_{back} are foreground and background covariances.

Next, we perform eigenvalue decomposition on $C_{contrastive}$ to compute the subspace V. The cancellation result is obtained by projecting the foreground signals into this contrastive subspace:

$$P_{clean} = P_{Fore} V \quad (10)$$

The projection operation selectively filters signal components common to both foreground (thoracic) and background (reference) regions by leveraging the eigen decomposition of the contrastive covariance matrix. The eigenvectors in V define a subspace that maximizes the variance ratio between cardiac signals and interference, effectively isolating heartbeat-specific activity while attenuating shared noise components. This spatial filtering preserves authentic cardiac waveforms by suppressing three key interference types: environmental noise from static objects, motion artifacts from respiration and body movements, and hardware-induced noise from the sensing platform.

With these refined components, as shown in Fig 5b, the thoracic region emerges more clearly and compactly, verifying the efficacy of our heart region refinement.

6.3. Iterative Stabilization of Heartbeat Signals

To handle temporal variations in environmental noise and human movement, 2FiA employs an iterative feedback mechanism that refines thoracic regions. After each round of reference region cancellation, we recompute the S-PSNR within the 3D space and prune any newly emerged reference region. Once the region stabilizes in both size and shape, we aggregate signals from the remaining pixels using Principal Component Analysis (PCA). PCA effectively extracts a representative waveform from the thoracic region input.

7. Separating Respiration and Heartbeat

Once the thoracic region is identified, 2FiA concentrates on isolating the true heartbeat signal. Even after identifying the thoracic region and applying the noise cancellation, heartbeat signals are still overshadowed by stronger co-located signals such as respiration as shown in Fig. 6.

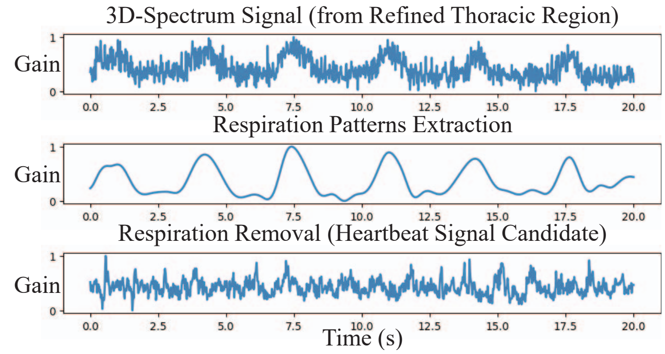


Figure 6: By adaptively estimating the noise energy ratio, 2FiA robustly extracts the true respiratory and heartbeat signals.

Respiratory signals are non-stationary signals, exhibiting variations in frequency over time [27]. As a result, they cannot be effectively handled using bandpass filtering methods [28], [29]. To overcome this, we employ an adaptive noise decomposition approach based on Ensemble Empirical Mode Decomposition (EEMD) [44], [45], breaking down the measured signal into IMFs. This adaptive approach effectively reduces mode mixing and clearly differentiates signal components.

IMFs Extraction: Inspired by EEMD, we incorporate multiple noise realizations into the measured signal before performing EEMD to avoid mode mixing. However, in contrast to standard EEMD, the added noise here is *adaptive*: its strength is tuned based on the relative power between the thoracic region and the corresponding reference signal component. This step yields a more robust separation of the IMFs. Formally, we generate multiple versions of the signal x extracted from the thoracic region by adding noise vectors N_i, giving $x_i = x + N_i$. After decomposing each x_i into IMFs, we compute a unique first residue $r_1 = x - \mathrm{IMF}_1$. Subsequent IMFs are then extracted by iteratively adding scaled noise terms:

$$\mathrm{IMF}_n = O_{n-1}(r_{n-1} + n_\sigma) \tag{11}$$

where n_σ) is a dynamic scaling noise, and $O_j(\cdot)$ represents the operator that extracts the j-th IMF from the signal, r_{n-1} is the residue from the $n-1$ iteration. Iterating this procedure prevents IMFs from blending respiratory information and heartbeat with environmental interference.

IMFs Selection: After obtaining the IMFs, 2FiA evaluates multiple combinations of IMFs systematically to accurately reconstruct the respiratory waveform. For each candidate reconstruction, we calculate the Breathing-to-Noise Ratio (BNR). The IMF combination that yields the highest BNR is selected as the optimal respiratory signal. After identifying the respiration-related IMFs and residual components, we isolate them from the full signal to suppress the dominant respiratory influence and associated noise. By reconstructing the respiration waveform from the selected IMFs and subtracting it from the original thoracic signal, we effectively eliminate large-amplitude, low-frequency variations. This subtraction unveils the underlying weak but consistent oscillations attributed to the heartbeat, yielding a cleaner and more distinguishable heartbeat signal candidate suitable for further verification and analysis.

The primary computational cost lies in IMFs extraction, with a complexity of $\mathcal{O}(T \log T)$ [44], where T denotes the length of signal samples. At a sampling rate of 50 Hz, with $T = 1500$ samples, with $R = 100$ noise realizations and $I = 10$ extracted IMFs. The total computational cost is approximately $R \cdot I \cdot \mathcal{O}(T \log T) \approx 1.58 \times 10^7$ operations. A low-power edge device such as the NVIDIA Jetson Nano (cost \$99) [46], equipped with an ARM Cortex-A57 CPU, offers up to 472G floating-point operations per second. It can solve this task in approximately $\frac{1.58 \times 10^7}{472 \times 10^9} = 0.03ms$, which is much shorter than the sampling duration.

8. RSA-Guided Heartbeat Verification

Given the subtle nature of heartbeat reflections and the potential for residual noise, we propose a verification approach that combines self-correlation and RSA-based features to detect and verify heartbeat signals from a candidate heartbeat signal.

RSA is a physiological phenomenon that reflects the natural variation in heart rate associated with the breathing cycle. It introduces a consistent, biologically driven relationship between respiration and heart rate, mediated by the autonomic nervous system, which regulates parasympathetic activity. Specifically, RSA creates a predictable pattern in heartbeat intervals: during inhalation, heartbeat intervals tend to shorten, while during exhalation, heartbeat intervals tend to lengthen.

To integrate RSA verification into our system, we adopt a three-stage approach that combines heartbeat cycles detection, respiratory phase detection, and RSA verification for real heartbeat in Algorithm 1. We first compute the autocorrelation sequence from the heartbeat signal to detect periodic patterns. Peaks in this sequence represent potential heartbeat intervals. We also segment the cleaned respiration cycle, as shown in Fig. 6, and identify distinct inhalation and exhalation phases. Finally, each detected heartbeat interval is evaluated using RSA constraints. A candidate heartbeat signal is valid if the change in heartbeat intervals matches the expected trend based on the current respiratory phase (i.e., increasing heart rate during inhalation or decreasing during exhalation). Otherwise, it is rejected as noise. This combined approach improves the robustness and physiological reliability of heartbeat detection.

This RSA verification strategy not only enhances the physiological plausibility of the extracted signal but also reinforces the spoofing resistance of our system. By leveraging involuntary cardiovascular-respiratory coupling, we introduce an additional barrier for adversaries attempting to mimic authentic biometric signals.

Algorithm 1 Heartbeat Signals Verification

Input: Respiration signals $Re(t)$, Heartbeat signals candidate $Hc(t)$
Output: Verified heartbeat $Hv(t)$
Step 1: Heartbeat Cycle Detection
Autocorrelation $C(t_1) = \sum_t^{N-t_1} H_c(t) \cdot H_c(t+t_1)$
$Hv(t) = \texttt{find\_peaks}(C(t_1))$
Each peak $p_i \in Hv(t)$ is considered a candidate heartbeat
Step 2: Respiration Phase Detection
Exhale: $\mathcal{E} = \texttt{find\_peaks}(Re(t))$
Inhale: $\mathcal{I} = \texttt{find\_peaks}(-Re(t))$
The phase $\mathcal{P}_{re}$ can be derived from $\mathcal{I}$ and $\mathcal{E}$
Step 3: RSA Verification for Real Heartbeat
for $i = 0$ to $|Hv|$ **do**
| $\Delta_{\text{prev}} = Hv[i] - Hv[i-1]$
| $\Delta_{\text{next}} = Hv[i+1] - Hv[i]$
if $\mathcal{P}_{re} ==$ inhale and $\Delta_{\text{next}} < \Delta_{\text{prev}}$ **then**
| $Hv[i]$ = True
else if $\mathcal{P}_{re} ==$ exhale and $\Delta_{\text{next}} > \Delta_{\text{prev}}$ **then**
| $Hv[i]$ = True
else
| $Hv[i]$ = False
end if
end for

9. Dual-Factor Authentication Model

9.1. Dual-Signal Feature Extraction

Our authentication system employs morphological features derived from reconstructed respiratory and heartbeat pattern to extract unique physiological characteristics. Specifically, we highlight heartbeat signals for fiducial-based feature due to their distinct and consistent peaks, enabling reliable identification of key landmarks within each cardiac cycle. In contrast, respiratory signals typically exhibit smoother, cyclic waveforms that lack consistently sharp peaks or clearly defined landmarks, thus for respiratory analysis, we adopt a non-fiducial feature strategy.

As shown in Fig. 7, fiducial features consist of identifiable points within heartbeat signals, notably characterized by distinct peaks and valleys. These points serve as consistent references, allowing precise extraction of physiological parameters and detailed pattern analysis. We then smooth and interpolate the signal for detecting fiducial points within each cycle and align the segments to a common length for further analysis.

To effectively capture respiratory signal characteristics, we perform waveform analysis on reconstructed respiratory data. We segment reconstructed respiratory signals into individual breathing cycles to facilitate meaningful feature extraction. Given the natural variability in breathing cycle durations, each segmented respiratory waveform is interpolated to match the length of the longest observed respiratory cycle within the dataset. This interpolation preserves the detailed morphological structure of the respiratory waveforms and ensures consistent comparative analyses across different cycles. This approach preserves the detailed respiratory waveform structure and facilitates comparison across different segments.

Additionally, we calculate the derivatives of these aligned heartbeat and respiratory waveforms to accentuate critical variations. Sepcifially, we computed derivatives for both the respiratory and heart-rate waveforms. Each derivative was calculated by taking the difference between adjacent points, giving two new waveforms. Derivative analysis highlights instantaneous changes in heartbeat and respiratory acceleration, capturing dynamic physiological features that enhance individual discrimination while simultaneously mitigating noise interference.

The combined morphological and derivative features extracted from heartbeat and respiratory analyses are integrated into a composite representation. All waveforms were plotted and combined into an image-like unified representation. This unified representation serves as the input to a Siamese Neural Network (SNN), specifically designed to leverage these detailed physiological characteristics for robust and reliable authentication.

9.2. Authentication via Siamese Embedding

For dual-factors authentication, we utilize a SNN trained with triplet loss [48]. The SNN effectively learns similarity

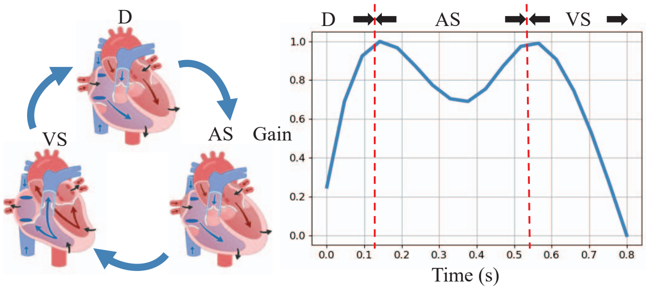


Figure 7: Illustration of cardiac motion and its corresponding validated heartbeat waveform extracted from 2FiA. D: Diastole, AS: Atrial Systole, VS: Ventricular Systole [47]

metrics between heartbeat and respiratory features, making it highly suitable for authentication tasks.

The network training process utilizes triplets of instances: an anchor (x_{a_i}), a positive sample (x_{p_i}), and a negative sample (x_{n_i}). The positive sample comes from the same individual as the anchor, while the negative sample comes from a different individual. These instances are processed through the same feedforward network F to generate their embedded representations. The training objective is to ensure the anchor's embedding is closer to the positive sample's embedding than to the negative sample's. This is achieved by minimizing the triplet loss:

$$L = \sum_{i=1}^{N} \left[|F(x_{a_i}) - F(x_{p_i})|_2^2 - |F(x_{a_i}) - F(x_{n_i})|_2^2 + \alpha \right]_+ \quad (12)$$

where α represents the margin enforced between positive and negative pairs [48]. This margin ensures sufficient separation between embeddings of different individuals while maintaining compact clusters for the same individual.

The network is trained to generate embeddings that satisfy two key criteria in order to improve efficiency: (1) heartbeat and respiratory patterns from the same individual should produce closely clustered embedding vectors, and (2) heartbeat and respiratory patterns from different individuals should result in widely separated embeddings. When authenticating a user, the system computes distances between their current embedding and all stored profile embeddings. The identity associated with the smallest distance is considered the closest match, enabling accurate user identification.

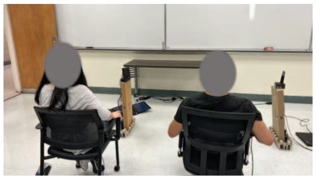
10. Performance Evaluation

10.1. Experimental Setups

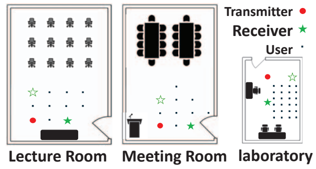
10.1.1. Devices. In our experimental setup, we deployment 2FiA using the commodity WiFi devices. Specifically, we utilize Dell LATITUDE laptops, each outfitted with Intel 5300 network interface cards (NICs), serving as both transmitters and receivers. The transmitter is configured with three antennas. On the receiver side, nine antennas are arranged in an L-shaped configuration, as illustrated in Fig. 8a. This L-shaped configuration consists of two orthogonal uniform linear subarrays. Each subarray is built using two NICs connected via a signal splitter, allowing them to share a central antenna. This shared antenna is used to synchronize phase shifts across the NICs. This configuration emulates potential antenna arrangements expected in future WiFi standards (e.g., WiFi 7 supports up to 16 antennas). During operation, the receivers run in monitor mode to passively capture transmissions from the sender. Both transmitting and receiving antennas are uniformly spaced at half-wavelength intervals (approximately 2.8 cm). We capture CSI data using the Linux 802.11 CSI Tool [49]. The WiFi signals are transmitted on a 5GHz channel with a 40MHz bandwidth, using 30 OFDM subcarriers, and the packet transmission rate is set to 1000 packets per second. Our system eliminates the need to collect ground-truth respiration or heartbeat waveforms for training the authentication model [50]. However, for evaluation purposes, we collect ground-truth respiration and heartbeat data from multiple users. Each participant is instructed to wear a commercial Neulog Respiration Belt and a Neulog Heart Rate & Pulse Logger Sensor [51], which is attached to their finger, to provide reliable reference signals for performance comparison.

10.1.2. Environment Settings. We conduct our experiments in three distinct indoor environments: a lecture room, a meeting room, and a laboratory, as illustrated in Fig. 8b. The lecture room contains multiple pairs of desks and chairs, a long lectern and measuring 6m $\times$ 9.5m. The lab room size is 4m $\times$ 6m. The meeting room includes two long tables, multiple chairs, and a projector. The room size is 6m $\times$ 9.5m. The laboratory room contains 8 labtop computers, desks, and chairs. Across all three locations, the desks and tables were wooden, while the chairs were made of metal and plastic. Each room has a different layout, size, and furniture such as chairs, tables, and devices. These environmental setups impact WiFi signal propagation. The various evaluation environments aim to assess system performance in diverse real-world environments. To evaluate our system, we recruited 36 participants, 20 male and 16 female, aged 19–54 years, with body weights ranging from 45–90 kg and heights from 150–198 cm. All participants were volunteers without any compensation, invited through seminars and workshops in which the study was presented and questions were addressed prior to enrollment. In addition, all participants were monitored in their natural state. Their respiration rates varied between 10 and 30 breaths per minute, while their heart rates ranged from 61 to 110 beats per minute. Over seven months, we collected around 50 million WiFi CSI packets to evaluate our system. All collected data were pseudonymized with random subject IDs, encrypted, and stored securely on password-protected, access-controlled research computers. The data collection was approved by the IRB of the authors' institution.

10.1.3. Baselines. **Cuvrb** [8] proposes a user verification system that extracts morphological features and fuzzy wavelet packet-based features from CSI amplitude. To ex-



(a) Two-user authentication experiment in the lecture room.



(b) Floor plans setup showing varied environment layouts.

Figure 8: Illustration of experimental setups.

tract these features, they remove noise with an EMD-based filter and select subcarriers based on periodicity activities.

Multisense [31] proposes a multi-person respiration monitoring system that divides the CSI signals from two antennas to eliminate time-varying phase offsets. Then they apply RobustICA to separate signals from multiple users, followed by an optimal matching algorithm to align and merge segmented respiration patterns.

For single- and multi-user scenarios, we selected Cuvrb [8] as the representative single-user baseline and Multisense [31] as the multi-user baseline. Other WiFi sensing-based authentication systems [2], [3], [4], [6], [8], [9], [10], [11] primarily target alternative modalities or employ techniques that are encompassed within Cuvrb and Multisense.

10.1.4. Model Settings. The input to our model is structured as a 2D matrix resembling an image, formed by combining multiple features. We adopt EfficientNet-B0 [52] as the backbone embedding model, replacing its final classification layer with a linear layer that outputs an embedding vector of size 512. We conducted an empirical study comparing six architectures: ResNet-18/34/50 and EfficientNet-B0/B1/B3. ResNet18 underperformed with an F1-score of 90%, whereas the other five achieved comparable performance with F1-scores between 96% and 98%. Among these, ResNet-34/50 and EfficientNet-B1/B3 required substantially more parameters, leading to higher computational costs. In contrast, EfficientNet-B0 delivered similar accuracy with considerably lower complexity. Therefore, we selected EfficientNet-B0 as the backbone model for our system. The model is trained using a batch size of 32, with triplet margin loss as the training objective. We employ the Adam optimizer with a learning rate set to 0.01. To measure the similarity between embeddings, we use Euclidean Distance. The deep learning framework is implemented in PyTorch, and training is conducted on an NVIDIA RTX 3090 GPU and Google Colab equipped with an NVIDIA A100 GPU.

10.1.5. User Authentication. For authentication, the system first captures and stores respiratory and heartbeat features extracted from the WiFi signals of each registered user to construct personalized user profiles. During authentication, a target individual remains stationary while the system continuously monitors their respiratory and heartbeat signals. The comparison is performed in the embedding space using Euclidean distance. Intuitively, if the monitored respiratory and heartbeat signal originates from the legitimate user,

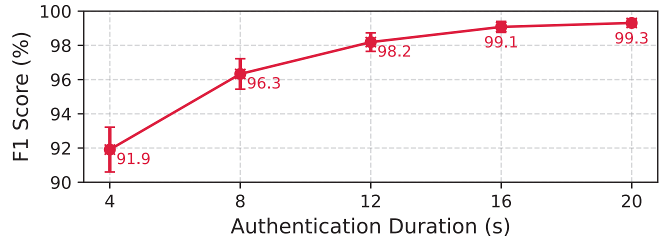


Figure 9: F1 scores of 2FiA under varying authentication durations.

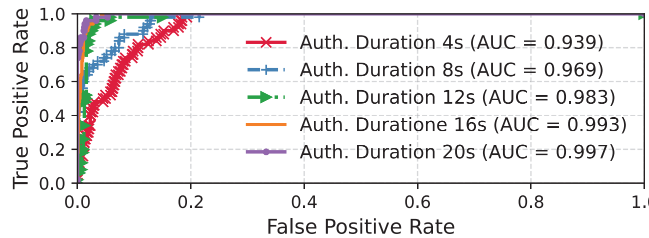


Figure 10: ROC curves and corresponding AUC values of 2FiA for different authentication durations.

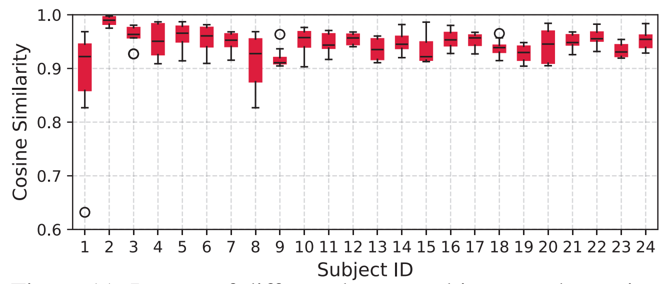


Figure 11: Impact of different human subjects on the cosine similarity of 2FiA respiration sensing.

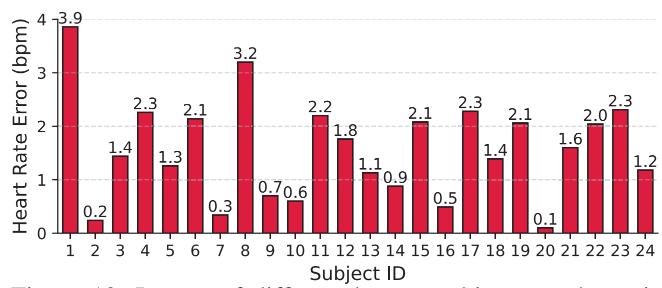


Figure 12: Impact of different human subjects on the estimation errors of heartbeat in 2FiA.

the distance between the extracted features and the stored profile should be small. In contrast, if the signal is from an unauthorized individual, the distance is expected to be significantly larger. To evaluate the system's ability to distinguish between users, we conducted experiments with dozens of participants. Each participant was designated as the registered owner in turn, while the remaining participants served as adversaries. A decision threshold was applied to differentiate between legitimate and unauthorized users based on the computed distances [53].

10.1.6. Evaluation Metrics. Authentication accuracy is evaluated using the F1 score, also known as the F-measure, which is non-sensitive to class distribution and can avoid misleading accuracy measurement when the true class distribution is unbalanced. The F1 score is the harmonic mean of precision and recall, providing a balanced metric that accounts for both false positives and false negatives. It is widely adopted for assessing classification performance in security-sensitive applications. The F1 score is defined as:

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2 * TP + FP + FN} \tag{13}$$

where the True Positive (TP) is 2FiA correctly identifies a legitimate (authorized) user. False Positive (FP) is 2FiA incorrectly classifies an unauthorized user as legitimate, potentially granting access to an attacker. False Negative (FN) is 2FiA incorrectly rejects a legitimate user, denying access to the rightful owner. F1-score incorporates both FP and FN rates.

Receiver operating characteristic (ROC) curve illustrates the trade-off between the true positive rate (TPR) and the false positive rate (FPR) at various decision thresholds. The Area Under the Curve (AUC) of ROC is a value that refers to the area under the ROC curve, which characterizes the system's ability to distinguish legitimate users from impostors.

We use cosine similarity to evaluate the correlation between the extracted respiration waveform and the ground truth respiration waveform. Cosine similarity measures the cosine of the angle between two vectors. Additionally, we report the heartbeat error rate in terms of Beats Per Minute (bpm), which quantifies the error number of cardiac cycles occurring within one minute occurring within one minute compared to the ground truth.

10.2. Overall Performance

In this section, we evaluate the overall performance of 2FiA across various environmental setups. During authentication, users position themselves within the sensing zone and remain stationary while the system captures their respiration and heartbeat signals for contact-free authentication. As shown in Fig. 9, 2FiA achieves F1 scores of 91.91%, 96.33%, 98.19%, 99.08%, and 99.31% with corresponding standard deviations of 1.31%, 0.89%, 0.54%, 0.32%, and 0.20% for authentication durations of 4s, 8s, 12s, 16s, and 20s, respectively. Fig. 10 further shows that the AUC improves from 93.9% to 99.7% over the same time intervals. These results demonstrate that longer authentication durations enhance both the accuracy and stability of 2FiA by allowing the system to extract more robust and distinguishable physiological patterns. The consistently high AUC values (all above 90%) indicate strong discrimination between legitimate users and adversaries.

Environment-independent futures evaluations: Furthermore, we not only evaluate the final authentication performance but also provide a detailed analysis of the extracted respiration and heartbeat signal. Evaluating these intermediate signals is essential, as the effectiveness of the authentication system fundamentally depends on the quality of the captured environment-independent patterns [5], [15]. In addition, this analysis improves the interpretability of our system, offering greater transparency compared to end-to-end black-box approaches. Specifically, we evaluate the cosine similarity between the extracted respiration waveform and the ground-truth respiration waveform to verify the fidelity of the extracted respiratory signal. As shown in Fig. 11, 2FiA consistently achieves cosine similarity above 90% across diverse users. This demonstrates 2FiA's ability to accurately and consistently extract respiration patterns across users. In contrast, heartbeat signals exhibit sharp peaks with short durations, not as smooth as the respiration waveform. As a result, even minor noise can significantly distort cosine similarity, making them unreliable for evaluating heartbeat accuracy. Thus, cosine similarity is unreliable for evaluation. Instead, we use heartbeat error rate in BPM, which more robustly reflects extraction accuracy. As shown in Fig. 12, 2FiA achieves an average heartbeat estimation error of just 1.5 bpm, demonstrating precise and reliable true heartbeat extraction.

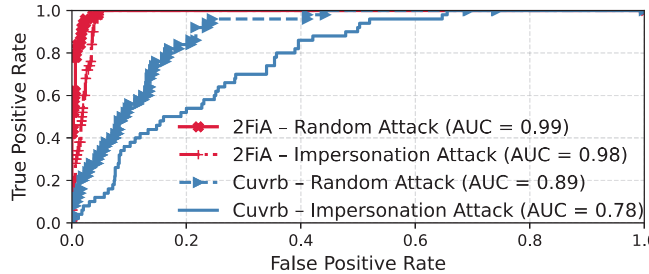


Figure 13: The ROC curves with AUC under random attack and impersonation attack compared to the state-of-the-art system.

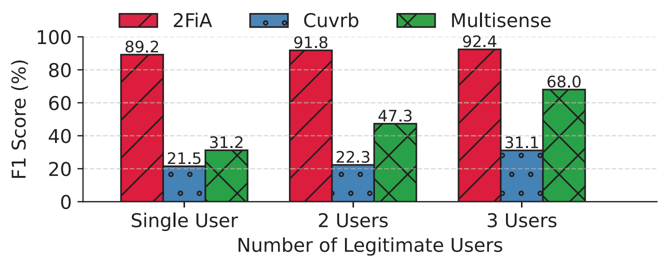


Figure 14: F1 scores of 2FiA, Cuvrb, and Multisense with varying numbers of legitimate users (1–3) in a shared environment with five individuals.

10.3. System Under Attacks Performance

To assess the robustness of 2FiA against adversarial attempts discussed in Section 2, we evaluate its performance under three types of attacks: random attack, impersonation attack, and replay attack. In a random attack, adversaries enter the sensing environment without any prior knowledge of legitimate users' respiration and heartbeat patterns. In contrast, impersonation attack assumes that the attacker mimics the victim's respiration behavior based on prior observations or recordings, as detailed in Section 2.2. In replay attack, the adversary captures and stores WiFi signals during a prior authentication session and later replays them in an attempt to bypass the authentication system, as detailed in Section 2.2.

We compare the performance of 2FiA with an existing state-of-the-art WiFi sensing-based authentication system, Cuvrb, which relies solely on respiration sensing. The ROC curve in Fig. 13 provides an intuitive understanding of system performance. Specifically, 2FiA achieves AUCs of 99% and 98% under random and impersonation attack scenarios, respectively. In contrast, Cuvrb achieves AUCs of only 89% and 78% under the same conditions. While Cuvrb fails to resist replay attack, our system can detect replay attack by analyzing environmental spectrum inconsistencies, as detailed in Fig. 3a. The environmental background signature in replayed signals differs significantly from that of the original session. Specifically, When the attacker device is positioned far from the receiver, it introduces environmental background discrepancies. In contrast, when the attacker moves very close to the receiver, the overall background appears more similar to the legitimate case. However, such close proximity introduces strong, device-induced multipath reflections between the attacker device and the receiver device, which are not present from the legitimate transmitter. Leveraging this discrepancy, our system detects replay attacks with over 95% by analyzing environmental consistency.

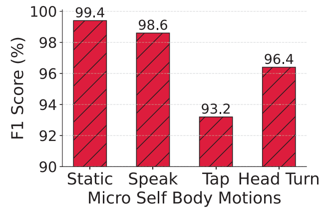
These results demonstrate that 2FiA is highly effective in distinguishing legitimate users from both types of impersonation attacks. Unlike respiration, which can be consciously modulated, heartbeat signals are governed by the autonomic nervous system and are therefore more difficult to spoof. While attackers may approximate respiratory patterns through careful observation and practice, impersonating heartbeat signals—particularly those driven by RSA—poses a substantial challenge. This physiological constraint introduces a critical layer of security, making 2FiA substantially more robust against impersonation attempts.

10.4. Multi-user Authentication Performance

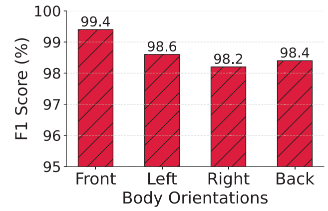
In this section, we evaluate the performance of 2FiA under varying numbers of co-existing users. In real-world environments such as healthcare facilities, secure workplaces, and smart homes, multiple users often coexist in the coverage area of a WiFi sensing system. However, most existing WiFi-based authentication systems assume single-user setting, which limits the applicability and robustness in shared spaces. Without the ability to differentiate and authenticate among co-existing users, these systems are susceptible to unauthorized access and signal contamination from nearby individuals.

We conduct experiments in a shared environment with five individuals, testing three configurations: one, two, and three legitimate users among the participants. In each scenario, the legitimate user(s) are authenticated, while the remaining individuals act as adversaries. This experimental setup simulates a real-world challenge where the system must reliably accept authorized users while rejecting unauthorized ones present in the environment. As shown in Fig. 14, 2FiA outperforms both baseline methods. 2FiA achieves high F1 scores of 89.2%, 91.8%, and 92.4% in the one-user, two-user, and three-user scenarios, respectively. Cuvrb achieves F1 scores of 21.5%, 22.3%, and 31.1%, respectively. Multisense achieves F1 scores of 31.2%, 47.3%, and 68.0%, respectively. All three systems are evaluated using the same dataset to ensure consistent comparison.

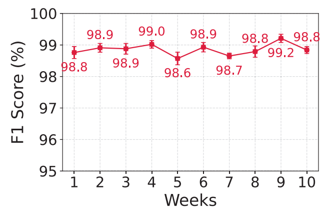
These results demonstrate that our system can authenticate multiple users by leveraging spatial information derived from WiFi signal propagation paths. This physical spatial information enables 2FiA to accurately localize and identify legitimate users. Multisense successfully separates users' signals using Independent Component Analysis (ICA), but it cannot determine which user is valid, limiting its utility in authentication scenarios. In contrast, 2FiA resolves this ambiguity through spatial information. The improved performance of Multisense over Cuvrb can be attributed to its ability to separate multiple users and its use of phase division between two antennas, which effectively mitigates time-varying phase offsets and enhances the clarity of respiration signals.



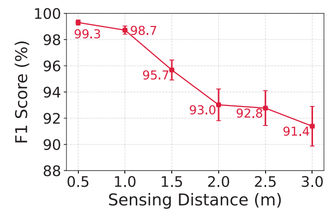
(a) Impact of micro self-motion on 2FiA authentication.



(b) Effect of user orientation on authentication accuracy.



(c) Authentication stability over a 10-week longitudinal study.



(d) Authentication performance across varying sensing distances.

Figure 15: Robustness evaluation of 2FiA under practical conditions.

10.5. Robustness Authentication Performance

To assess the robustness of 2FiA under realistic and diverse deployment conditions, we evaluate its performance across four key dimensions: micro self-motions, user orientations, longitudinal stability, and sensing distance.

10.5.1. Impact of Micro Self-Motions. In this section, we evaluate the performance of 2FiA in the presence of common micro self-body motions that users may exhibit unconsciously during authentication. Micro motions introduce minor perturbations to the WiFi signals, potentially affecting the quality of physiological signal extraction and, consequently, the reliability of authentication. We conduct experiments under four conditions: static, speaking, tapping, and head turning. In each case, the subject remains in the authentication zone while performing the designated action.

As shown in Fig. 15a, 2FiA maintains high authentication performance, achieving F1 scores of 99.4%, 98.6%, 93.2%, and 96.4% under the respective scenarios. The results indicate that speaking and head turning have minimal impact on performance, while tapping causes a slight degradation due to its localized upper-body movement, which can interfere with subtle heartbeat signal extraction. These findings demonstrate the robustness of 2FiA against natural micro self-motions, highlighting its practical applicability in dynamic user environments.

10.5.2. Impact of Body Orientations. In this section, we evaluate the performance of 2FiA under four different body orientations. In realistic environments, users may not always face the sensing device directly during the authentication process. Variations in body orientation can alter WiFi signal propagation [54] and potentially reduce the clarity of the captured physiological signals.

We conduct experiments in four orientation settings: facing the receiver, turned left, turned right, and back facing the receiver. As shown in Fig. 15b, 2FiA maintains strong performance across all directions, achieving F1 scores of 99.4%, 98.6%, 98.2%, and 98.4%, respectively. A slight performance drop is observed when the user turns their back to the device, primarily due to respiration signal obstruction. These results demonstrate that 2FiA is resilient to variations in body orientation, making it suitable for practical deployments where users may not always maintain a consistent posture relative to the sensing hardware.

10.5.3. Impact of Longitudinal Study. In this section, we evaluate the longitudinal robustness of 2FiA. This evaluation is essential to ensure that the system maintains consistent accuracy over time without requiring frequent re-enrollment. We conduct a longitudinal study to assess authentication performance over a ten-week period. The user undergoes the same authentication protocol under controlled conditions each week.

As shown in Fig. 15c, 2FiA achieves consistently high F1 scores ranging from 98.57% to 99.21%, with standard deviations remaining below 0.20% across all weeks. These minimal fluctuations demonstrate that the extracted respiration and heartbeat features remain stable and discriminative over time.

10.5.4. Impact of Sensing Distance. In this section, we evaluate the authentication performance of 2FiA across varying user-to-device distances. This experiment reflects real-world deployment scenarios where users may not always be positioned at a fixed location relative to the WiFi transceivers. Variations in distance can alter the signal-to-noise ratio, multipath profile, and reflection intensity, all of which may affect the system's authentication capability.

We conduct experiments at six distances: 0.5 m, 1 m, 1.5 m, 2 m, 2.5 m, and 3 m. As shown in Fig. 15d, 2FiA maintains high F1 scores above 95% up to 1.5 m, with peak performance of 99.3% at 0.5 m. Beyond this range, performance gradually declines, reaching 91.39% at 3 m. As physiological signal strength weakens with increasing distance, it becomes more susceptible to environmental noise. Although noise suppression techniques are applied, subtle motions such as heartbeat remain sensitive to distance-related signal degradation. In contrast, respiration signals remain reliably detectable even at longer distances.

Despite the observed performance degradation, 2FiA continues to operate effectively across all tested distances, demonstrating its practicality for contactless authentication.

10.6. Ablation Study

In this section, we present an ablation study to separately verify respiration and heartbeat feature sets. Both feature sets were derived from the same WiFi dataset and fed into the same SNN for training and evaluation. The results show that respiration-only-based authentication achieved an F1-score of 90.3% , with a false positive rate of 3.1% and a false negative rate of 5.0%. Heartbeat-only-based authentication achieved an F1-score of 94.2%, with a false positive rate of 1.8% and a false negative rate of 3.0%. When combining both features, it achieves an F1-score of 98.2%, with a false positive rate of 0.4% and a false negative rate of 1.5%. These results show that heartbeat-only features outperform respiration-only features. Combining both further improves performance.

11. Related Work

WiFi sensing has demonstrated broad applicability across various domains. In response to the growing momentum in this field, the IEEE 802.11bf Task Group was established to standardize WiFi sensing capabilities, aiming to support sensing-based services such as activity recognition [55], [56], [57], [58], gesture recognition [59], [60], [61], and physiological signals sensing [28], [31], [62]. These sensing services open up new opportunities for enhancing security and privacy by reducing intrusiveness, offering a clear advantage over contact-based sensor devices [50], [63].

WiFi Sensing Authentication Systems. Recent advances in biometric identification have led to the development of diverse approaches that address both accuracy and security in complex environments. For instance, XModalID [2] introduces a WiFi-video cross-modal gait-based person identification system, thereby enhancing recognition in real-world scenarios. Meanwhile, AutoID [3] employs convex tensor shapelet learning to enhance human identification from WiFi data, highlighting the potential of advanced machine learning methods in signal processing contexts. 3D-ID [4] extend this paradigm into three-dimensional space with a WiFi vision-based approach, enabling refined person re-identification by extracting spatial features from wireless signals. Building on these identification methods, MultiAuth [10] facilitates multi-user authentication using a single commodity WiFi device, while their later work on Fingerpass [6] showcases continuous user authentication via finger gesture recognition tailored for smart home environments. Complementarily, Shi et al. explore smart user authentication that leverages the actuation of daily activities through WiFi-enabled IoT devices, underscoring the value of integrating behavioral context into authentication frameworks [11]. Liu et al. introduce a novel continuous user verification system based on respiratory biometrics, further expanding the range of non-intrusive, passive sensing approaches [8]. Along this line, BreathID [9] further advances respiration-based identification by proposing a CSI ratio with a weighted multidimensional dynamic time warping algorithm to realize the human identification. Collectively, these studies illustrate a robust evolution of WiFi-based biometric systems, from challenging through-wall identification to continuous and context-aware authentication, laying a strong foundation for future research in this dynamic field.

However, existing WiFi-based authentication systems mostly rely on voluntary or semi-voluntary biometric signals. While these signals can effectively capture user-specific behavioral features, they remain relatively vulnerable to spoofing due to their observable and reproducible nature.

Authentication Leveraging Heartbeat. Recent research in heartbeat-based authentication has explored a range of techniques that utilize unique heartbeat signals to verify user identity in a secure and ongoing manner. To monitor heartbeat, common sensing modalities include electrocardiography (ECG), seismocardiography (SCG), photoplethysmography (PPG), and phonocardiography (PCG), each capturing cardiac activity through electrical, mechanical, optical, or acoustic signals, respectively. ECG-based authentication laid the foundation by demonstrating that electrocardiogram signals can serve as reliable biometric identifiers [64], leveraging the distinct electrical activity of the heart. Cardiac Scan [26] extended this idea by introducing a Doppler radar that continuously monitors heart signals for user authentication, enhancing convenience while maintaining security. Unlock with Heartbeat [65] further advanced this paradigm by integrating heartbeat-based authentication into mobile phones, allowing users to unlock devices using inherent cardiac patterns. CardioCam [66] utilized mobile cameras to capture subtle heartbeat-induced color changes in facial skin for continuous user verification. More recently, Heartprint [67] introduced an in-ear microphone-based method that passively captures heart sounds for unobtrusive authentication. Collectively, these systems highlight the growing potential of heartbeat-based authentication to provide robust, passive, and user-friendly security across diverse platforms.

However, these approaches typically rely on dedicated sensors or constrained user positioning, which may limit their applicability in ambient or device-free environments. In contrast, WiFi sensing enables contactless and infrastructure-independent authentication, offering greater flexibility for unobtrusive deployment.

12. Conclusion

In this paper, we presented 2FiA, the first two-factor WiFi sensing-based authentication system that integrates semi-voluntary respiration and involuntary heartbeat signals to enhance impersonation resistance and authentication robustness. Leveraging WiFi CSI, 2FiA localizes thoracic region using 3D spatial profiling and extracts fine-grained physiological signals through adaptive noise decomposition and RSA-based heartbeat validation. These signals are then encoded into a unified feature embedding and processed via an SNN for identity verification. Extensive experiments across diverse scenarios, including adversarial impersonation, multi-user environments, and long-term evaluations, demonstrate that 2FiA achieves high authentication accuracy and strong resilience under realistic deployment conditions. The incorporation of involuntary physiological traits enhances the system's security and offers potential for the next generation of seamless and privacy-preserving authentication systems.

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Appendix A. Meta-Review

The following meta-review was prepared by the program committee for the 2026 IEEE Symposium on Security and Privacy (S&P) as part of the review process as detailed in the call for papers.

A.1. Summary

The paper proposes a WiFi sensing based approach for user authentication. WiFi channel information, captured by a receiver device, is used to extract a user's heart and breathing rates, which are then used for authenticating the user. This approach requires locating the WiFi signals that were reflected by the heart area of the user to the receiver device, removing noise, extracting the respiration rate, extracting the heart rate, which is difficult due to the weak signal so some additional processing relying on correlations between the heart and breathing rates is needed, and extracting features. The features are used in a Siamese network to compare a user's measured template to the user's registered template.

A.2. Scientific Contributions

- Provides a Valuable Step Forward in an Established Field
- Addresses a Long-Known Issue

A.3. Reasons for Acceptance

1) Overall design: The idea of combining respiratory data with heart beat data while also leveraging their relationship is interesting. The ability of the approach to extract accurate heart beat data from WiFi signals (as opposed to dedicated wireless methods) is impressive and might have other applications beyond authentication and security.
2) Implementation and evaluation: The paper provides a prototype implementation with a thorough evaluation. The experiments were conducted over seven months across three distinct indoor environments (lecture room, meeting room, and laboratory), with longitudinal stability over 10 weeks. The evaluation also looks at mimicry and replay attacks and multi-user scenarios.

A.4. Noteworthy Concerns

1) Section 8 proposes an optional approach for verifying the heartbeat but the paper does not state how much this approach helps.